

Analog Smart Edge Detector Solution

Background

The project my senior capstone group undertook was to develop a direction of arrival (DoA) system for an underwater autonomous drone that would be used by our school's robotics club in the [Robosub](#) Competition. The main part of the project therefore was to create a receiver system that would receive and process a signal, before running the [MUSIC](#) algorithm to determine the DoA for the incoming signal.

The MUSIC algorithm principally requires that the background noise of our environment be completely uncorrelated with the signal received. However, our system is meant to be in a pool, meaning that echoes of the transmitted signal are inevitable. Once the echoes arrive, our data would no longer be useful as the system begins outputting incorrect and unstable results.

It is therefore important to record as much data as possible before the echos arrive, but we quickly ran into two issues. First, we had chosen to use an STM32's onboard Multiplexed ADC, which meant that for our four channel system we were sampling at 400 kbps, and due to our testing configuration, echoes were arriving after 300 μ s, leaving us only able to get 120 good samples. Second, initial testing was done with commercial pingers that were insensitive enough to not output random noise spikes due to the pool. However, the final receivers that would be used in the Robosub competition are hand-made and not adequate enough at rejecting noise, leading to excessive amounts of voltage spikes both in quantity and magnitude. These can be seen in figures 1 and 2 which show two types of noise

recorded right before a signal: figure 1 shows a large (greater in magnitude than the actual signal) and rapid spike, while figure 2 shows background noise that sustains for slightly longer.

To get around this we implemented digital solutions to determine if the receivers were outputting noise or our signal, and then the STM would record data, however, these solutions would take roughly 200 μ s, leaving only 100 μ s of good data, or roughly 40 samples. This is enough to measure a couple periods of our signal, however, the final DoA value was less accurate and much less consistent.

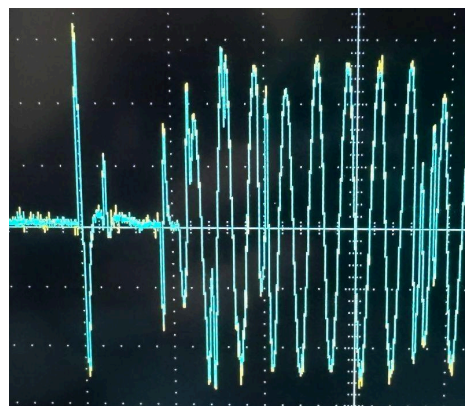


Figure 1: Large noise spike right before the transmitted signal is received.

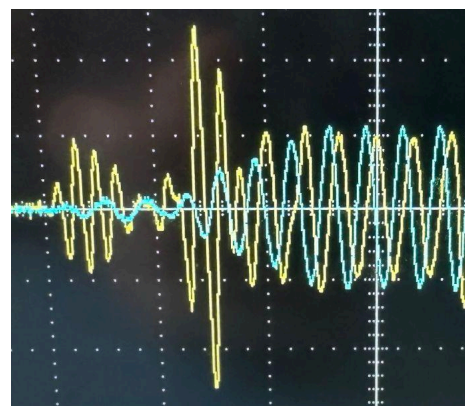


Figure 2: Sustained noise spike (in yellow), blue shows a low-pass filter applied.

Implementation

Theory of Operation

There are two important traits this signal detector must have: it needs to reliably determine the difference between signal and noise, and do so quickly (ideally less than 100 μs). As a bonus, the circuit should be as small as possible in order to fit on the current Low Noise Amplifier board.

To differentiate signal from noise the circuit observes the shape of the incoming signal as this makes it easy to determine the width of the incoming signal. The width is convenient to measure as whether noise is a spike or a small oscillation it will have a maximum duration it exists, in the order of microseconds. Meanwhile, the transmitted signal is broadcasted for 10ms. *Therefore, the circuit sets a threshold of time after which it will trigger if the signal is still persisting as that can only happen if the signal being measured is the desired transmitted signal.*

The first part of the circuit needs to extract the shape of the signal and this is done using an envelope detector, like the one seen in figure 3a. The output of an envelope detector, as shown in figure 3b can measure an approximately DC value for the duration of a signal.

The second part of the circuit needs to measure the duration of the signal based on its shape, and the envelope detector can actually be tuned to do this as well. By tuning the time constant for the RC-circuit on the output, the output voltage can be set to reach a threshold value of a predetermined number of oscillations. For instance, if the maximum number of oscillations ambient noise has been observed to make it around 6 (which is true from our test results), then setting the threshold to trigger after 10 periods means only the transmitted signal could do this. Also, the RC-circuit can act as a lowpass filter to reduce the influence of signals past our desired frequency of 40 kHz.

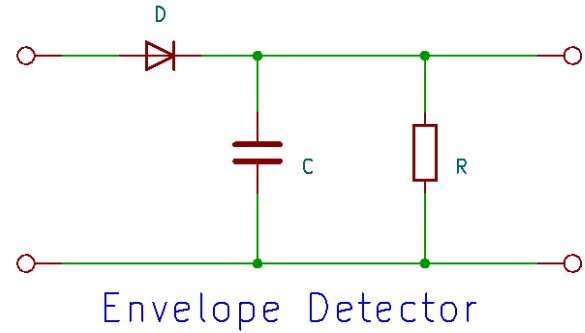


Figure 3a: Envelope Detector Circuit

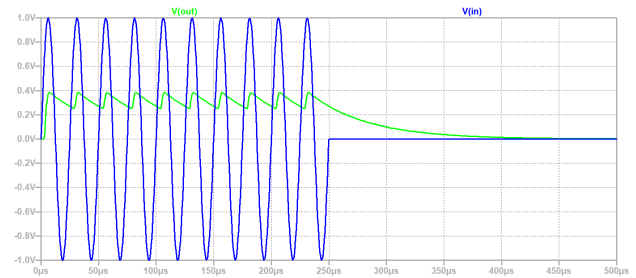


Figure 3b: Output of envelope detector circuit.

In order for the circuit to continually gain charge, it has to be tuned to gain more charge than it loses during each cycle. The envelope detector for this circuit works by rectifying the incoming signal so it is zero during the negative phase. During the positive half of each period the capacitor stores charge, and then discharges it through the resistor during the second half of the period. Calculating the condition for which more charge is gained than loss comes down to approximately satisfying the following condition: $\frac{2}{f} > RC$. For a frequency of 40 kHz, that means our time constant must be less than 50 μs .

One issue does arise from trying to calculate how much voltage is gained during each phase, which is that the amount gained depends on both the RC-time constant as well as the amplitude of the incoming signal. The incoming signal will vary between values of 200mV and 2V peak, which is too large of a swing such that one RC-time constant can

charge the capacitor a meaningful enough amount to reliably trigger only on the real signal in a time of 100 μ s or less. Therefore in order to move forward the circuit has to be augmented to normalize the amplitude of the rectified signal.

Limiter Circuit

A very simple method to normalize the rectified signal's amplitude is to limit it using a zener diode in the negative feedback loop of an op amp. In the essence of saving space and cost, since our incoming signal will be rectified, we only need one of the zener's in the feedback loop for our circuit. The two resistors need to be configured to set the gain in order to do two things: (1) such that any incoming signal will be amplified enough to force the zener into its breakdown region (but not exceeding it into the avalanche region), and (2) small enough such that the initial voltage on the output of the op amp is low enough to not bias the zener in its breakdown region.

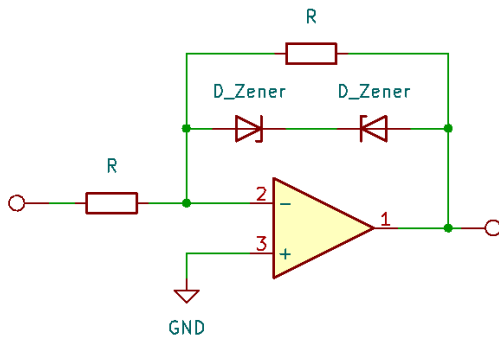


Figure 4: Basic Op Amp Limiter Circuit.

The first condition is necessary to make sure that the incoming signal is limited and therefore normalized to the zener diode's breakdown voltage, which in this application is either 2.4V or 2.7V. The second condition is necessary to ensure that the capacitor in the RC network starts at an initially lower voltage than the 2.4V or 2.7V. This allows the capacitor to charge up a meaningful difference in voltage over several oscillations. Figure 5 shows an

LTSpice simulation (the circuit under simulation is shown in figure 6) varying the gain of the op amp limiter stage from 10x (blue) to 100x (grey). As the gain increases, ripple on the output is drastically reduced, but the initial voltage that will be applied to the capacitor increases. Ripple is caused when the rectified signal drops to a value below which the gain is not sufficient to cause the zener to be biased in its breakdown region, which is seen clearly in the equation for the voltage across the zener: $V_z = (-1 - \frac{R_2}{R_1})v_i$.

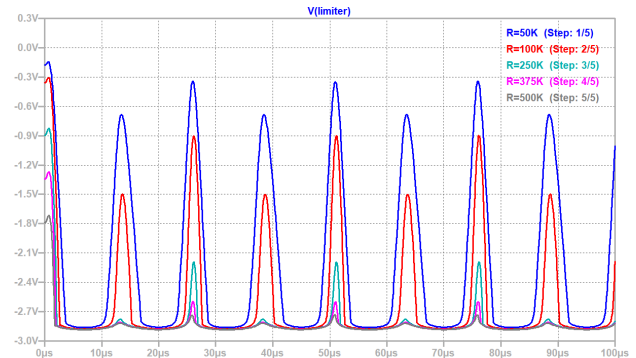


Figure 5: Output of limiter circuit for gain ratios from 10x to 100x.

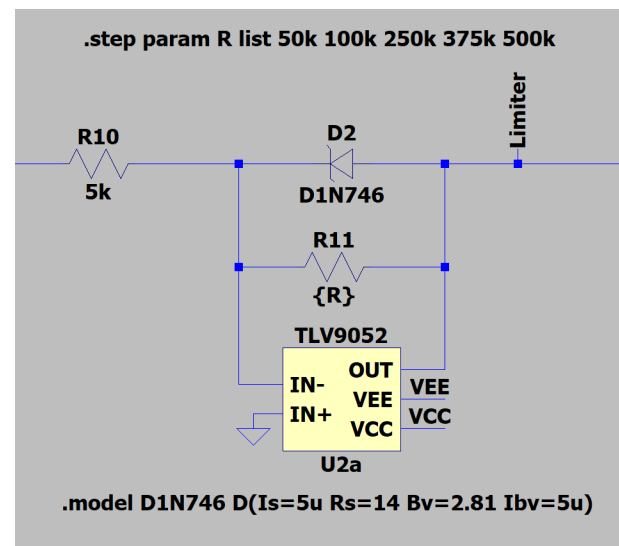


Figure 6: LTSpice circuit used to simulate the graph shown in figure 5. The TLV9052 SPICE macro model used was downloaded from Texas Instruments's website.

Precision is not necessarily important in this application, and figure 7 shows that despite the ripple the circuit still charges the capacitor appropriately. Additionally, a 200mV signal only needs a gain of 13.5x to reach 2.7V, meaning for lower gain, there is a larger number of signal amplitudes that are simply rejected because they do not cause the circuit to reach its limiting state, meaning they do not sufficiently charge the capacitor fast enough to incrementally add more voltage at the output. Therefore, lowering the gain adds an additional buffer to noise spikes of slightly lower magnitude. Figure 8 shows the (full-wave) rectified signal, the limiter output, and the capacitor voltage for a gain of 20x.

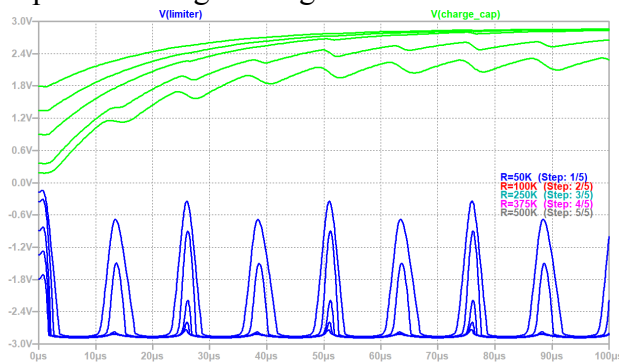


Figure 7: Voltage at output of limiter circuit and across the charge capacitor for various gain ratios.

Simulations show that a gain of 40x is needed, however, to ensure 200mV signals saturate the capacitor to 2.7V.

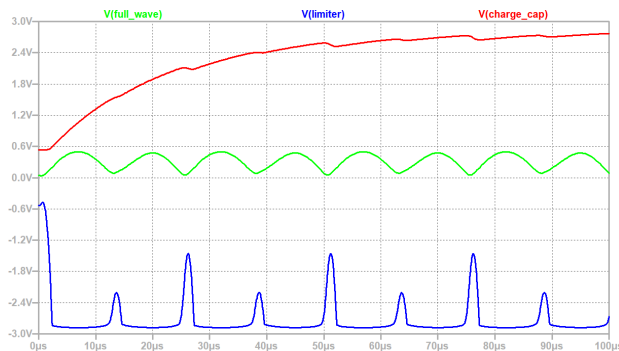


Figure 8: Output voltages of each stage with a resistor ratio of 40:1.

Low Pass Filter

The RC-network from a simple envelope detector can be upgraded to an active single-pole RC lowpass filter, and by doing so the RC network is isolated better from the previous and next stages' impedances. It also helps with drive capability for the next stage since otherwise the output would have to load the capacitor in parallel.

A single TLV9052 dual op amp chip can implement both the limiter and low pass filter circuits. The TLV9052 was chosen because it is inexpensive, low cost, relatively low-noise and low-current, and has a high slew rate and GBWP so it can handle the 2.7V peak-to-peak AC signal. The slew rate and GBWP are especially helpful as mentioned for the larger peak voltage swings but also to ensure the limiter circuit can latch to the zener breakdown voltage as quickly as possible.

Precision Rectifier

Another necessary improvement to the circuit is needed in the rectification stage. The input signal may be smaller than 600 mV so a precision rectifier is needed to compensate for the forward voltage drop of a normal silicon diode. Additionally, to halve the amount of time needed to charge the capacitor, a full-wave rectifier can be implemented.

There seem to be two main ways to implement a full-wave precision rectifier. Method one is shown in figure 9, and it comes from an article by Analog Devices [1]. As highlighted by the article this design has some inherent issues with needing matched precision parts, cross-over distortion with the diodes, and needing a dual power supply. Since this detector circuit will go on an LNA board that already has +/- 5V supplies, the last point is not important in our case.

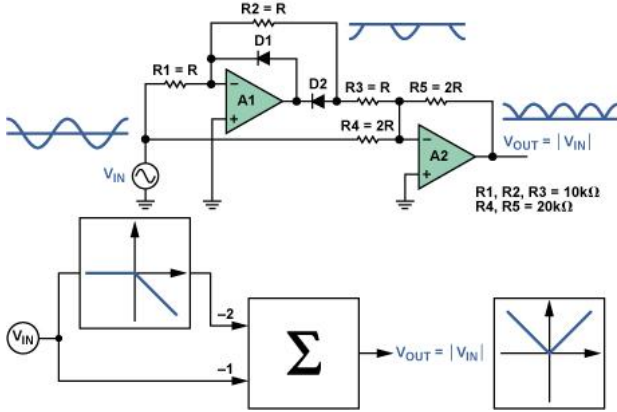


Figure 9: Precision Rectifier (Absolute value Circuit from Analog Devices)

A second design for a precision rectifier shown in figure 10 is from a Burr-Brown application note [2]. It has the benefit of only needing two matched resistors, but capacitor C1 must be matched well to resistor R1 to ensure A2 is stable in the feedback loop.

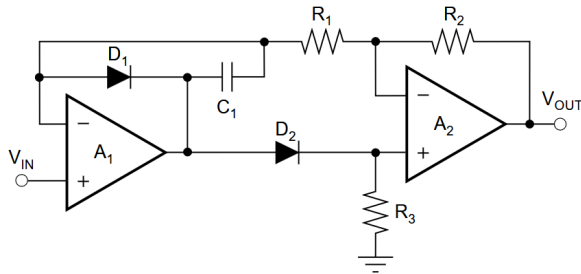


Figure 10: Precision rectifier from Burr-Brown Application Note.

In both cases, a monolithic difference amplifier IC can be used to account for the matched resistors needed, saving on components, cost, and reducing mismatch errors. Going a step further, however, the ADI article highlights how the AD8277 dual difference amplifier can perform the absolute value math function with no extra components. This is due to its ability to take negative values on the input and function with minimal distortion or errors, despite being a single-supply op amp. An LTSpice simulation in figure 11 showed that the output distortion scaled inversely to the input impedance of the next stage, however,

reductions in distortion were only marginally better after a load impedance of 10 kΩ, and given that the resistor ratio of the next stage's op amp is 20:1, it does not seem necessary to go higher.

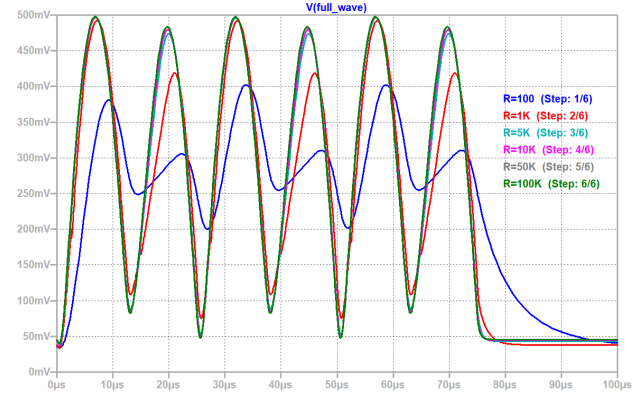


Figure 11: Output of AD8277 (two AD8276 SPICE Macro models from ADI website), for various input resistance values of the limiter stage.

Comparator

The final stage of the circuit is a schmitt trigger that will read when the charge capacitor has reached a certain threshold and then it will latch low. The output will go to an STM32 microcontroller GPIO pin that has an interrupt waiting for the pin to be pulled low.

The trigger is meant to latch after a certain duration of time. In order to measure time the amount of voltage added to the charge capacitor by each incremental oscillation must be calculated, and its value comes out to:

$$\Delta V_{\text{per period}} = V_m \left(1 - \frac{4\pi}{RC} f_{\text{full-wave}} \right).$$

Each period of our full-wave rectified signal is 12.5 μs and the circuit will latch after 8 oscillations for a total time of 100 μs. To calculate the RC time-constant needed, pick an upper threshold for the schmitt trigger, less than

2.7V, and set that equal to $8\Delta V_{\text{per period}}$. This yields one equation for RC and three conditions:

$$(1) \quad \tau = RC = \frac{1}{\pi f_{\text{full-wave}}} \frac{1}{1 - \frac{V_T}{8V_M}}$$

$$(2) \quad 1 - \frac{1}{\pi f_{\text{full-wave}} RC} > 0,$$

$$(3) \quad 1 - \frac{V_T}{8V_M} > 0,$$

$$(4) \quad 0 < V_T < 2.7V.$$

Condition 3 says that V_T is determined by the minimum input signal expected. Given that $(V_M)_{\min} = 200 \text{ mV}$, the trigger threshold should be less than 1.6V, and I will pick 1.5V. Given, $f_{\text{full-wave}}$ equals 80 kHz, equation 1 can be used to calculate $\tau = 4.276 \mu\text{s}$. Lastly, condition 2 states that τ must be greater than $3.98 \mu\text{s}$, which it is. A 470 pF capacitor and $10 \text{ k}\Omega$ (1%) resistor gives a time constant of $4.7 \mu\text{s}$.

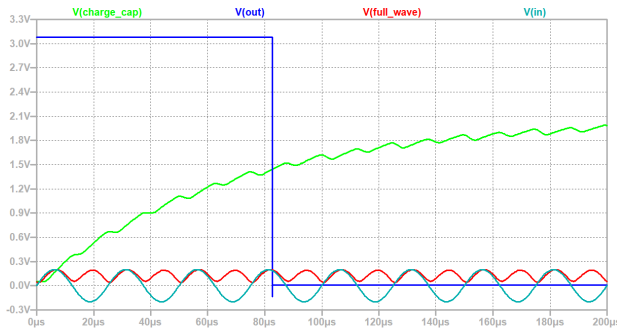


Figure 12: Full circuit operation. 200 mV input signal triggers a latch down after 3.5 periods or about $80 \mu\text{s}$.

The comparator is a TLV3201 and uses hysteresis. It is based on the reference circuit in the ICs datasheet, shown in figure 13 [3].

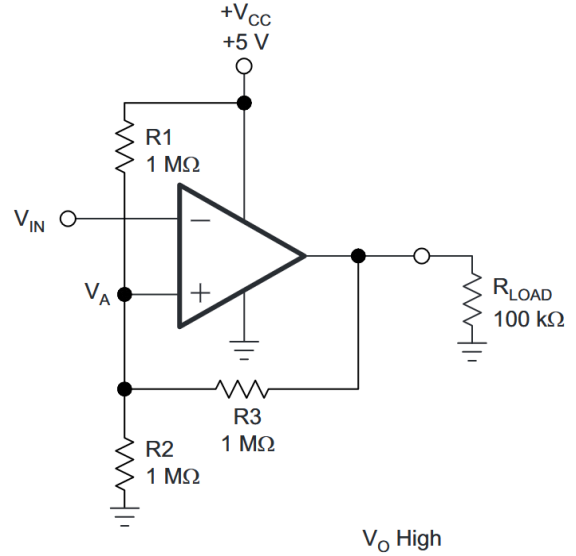


Figure 13: Schmitt trigger reference design, the resistor values are not accurate to the final design.

I used Excel's solver to simultaneously solve the characteristic equations given in the data sheet for setting the threshold voltages, as seen in figure 14. The values underneath the solved for resistor values are the E96 equivalents.

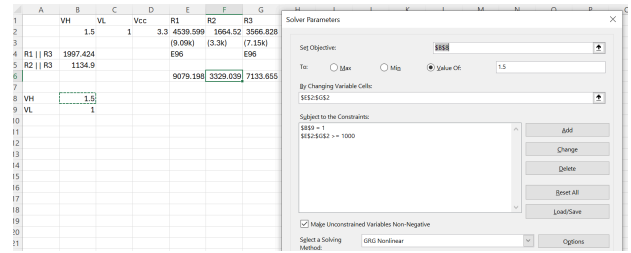


Figure 14: Excel sheet used to find threshold resistor values.

After some more fine tuning of values to ensure that signals down to 200 mV peak and up to 2V peak latch down in the time period between $70 \mu\text{s}$ and $100 \mu\text{s}$, shown in figures 15 and 16, the circuit values are shown in figure 17.

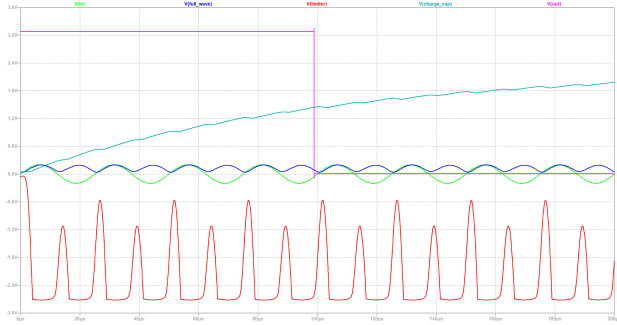


Figure 15: 200 mV peak input. Green is the input signal, dark blue is the rectified signal, red is the limited signal, blue is the capacitor charge, and pink is the output of the comparator.

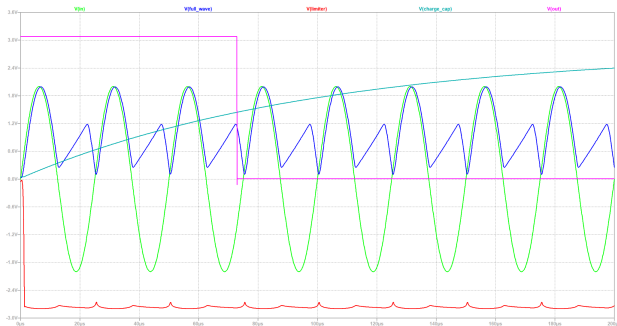


Figure 16: 2 V peak input.

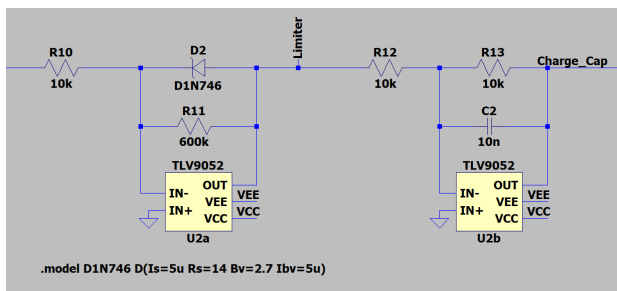


Figure 17: Op amp limiter and RC lowpass circuit LTSpice values.

PCB Implementation

The first implementation of this circuit, REV A, was slightly different from the current circuit that has been discussed so far. It follows

all the same principles, but instead of using the AD8277 for the rectifier, it used a TL072 op amp and an AD8276, meaning it also needed additional diodes and resistors. Additionally, the comparator before was a TL431 acting as a shunt voltage regulator for the non-inverting input of an LM311 comparator, as opposed to just using the TLV3201. These new changes cut down the design by 9 passive components (two of which were matched 0.1% resistors) and 2 ICs. This will help to shrink the PCB size and cost a fair amount.

The layout for REV A of the circuit board was rather straight forward. Extra care was taken at the input of the AD8276 because the circuit configuration for the rectifier had the TL072's output feeding the non-inverting input of the AD8276 making it a very high impedance node. Although the rest of the top layer is a copper fill stitched to the ground plane, the area around this high impedance node remained FR4 fiberglass to reduce parasitic capacitance.

Aside from the main circuit, had mounts for banana clips to supply $\pm 5V$ and ground, with a large electrolytics and two ceramic capacitors on each power line to decouple power supply noise. Additionally, every IC had an associated decoupling capacitor for their power pin. The passive components are all 0603 imperial unit SMD parts. Since it is a development board to test the circuit extra test pads were added.

The board for REV A was made with two layers because it is a relatively simple circuit, however, I could not find a necessarily elegant way to route the dual power supplies due to the test points and extra passive components, so for REV B it will be a four layer board.

JST connectors were used for input and output signals because the signal frequency is relatively low and the JST connectors and cables are very inexpensive compared to the BNC connector equivalents I could find online. The total cost of parts for the board comes out to \$15, and the boards themselves were \$5.

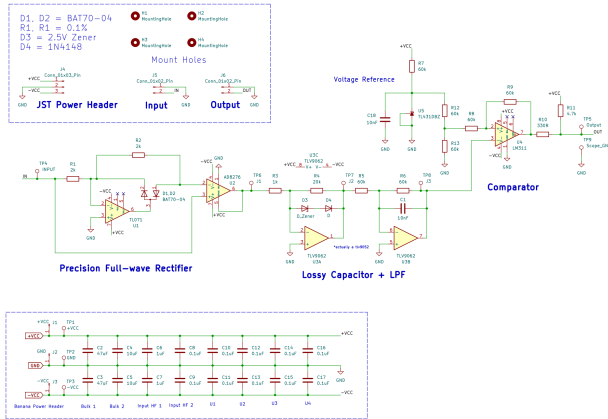


Figure 18: Schematic of REV A board.

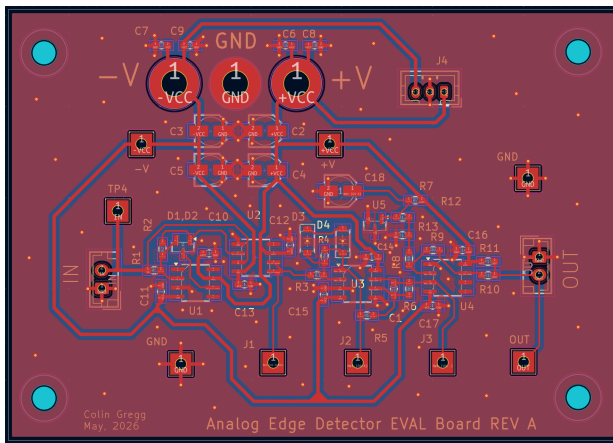


Figure 18: Layout of REV A board.

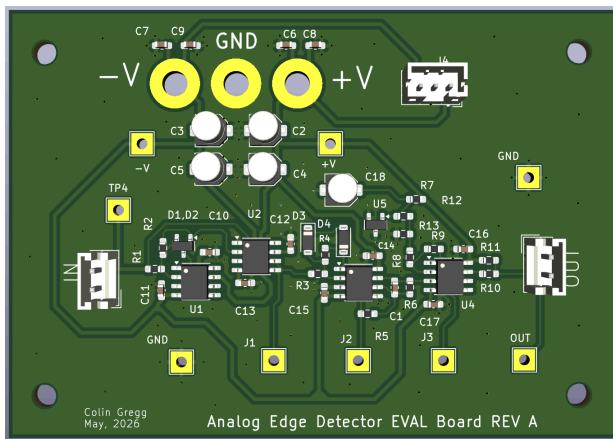


Figure 19: 3D view of REV A board.

Next steps

I made this board for my capstone project, however, I started in the last week of class and by the time I finished and ordered the board, even with express shipping, it would not have arrived until the day I had to move out of my apartment. I do not currently have test/lab equipment at my house and so have not been able to test the REV A board. I plan to remake the PCB with REV B implementing the circuit described in the other part of this report, and will hopefully have equipment to test it in September, 2026.

Sources

1. <https://www.analog.com/en/resources/analog-dialogue/articles/more-value-from-absolute-value-circuit-diff-amp.html>
2. <https://www.ti.com/lit/an/sboa068/sboa068.pdf?ts=1780815917420>
3. https://www.ti.com/lit/ds/symlink/tlv3201.pdf?ts=1780822756897&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FTLV3201